

Highlights

A State-Machine Stackelberg Game for Diagnosis–Repair in Natural-Language Optimization

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- Diagnosis–repair is cast as a Stackelberg inspection game defined on a LangGraph state machine as the stage-game apparatus.
- The diagnosis agent is an inspector that commits, first-mover, to a causal-order (data $\triangleright$ model $\triangleright$ code) probing policy with executed refutation.
- Accused agents are inspectees with a genuine deflection incentive; the inspector’s commitment induces truthful attribution.
- A proposition links causal-order inspection to a strictly smaller viable root-cause set than single-judge diagnosis.

A State-Machine Stackelberg Game for Diagnosis–Repair in Natural-Language Optimization

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ABSTRACT

We recast the error diagnosis and repair stage of an LLM multi-agent optimization pipeline as a Stackelberg *inspection game* defined on a state machine. A LangGraph finite-state machine orchestrates role-specialized agents (Data Engineer → Model Expert → Python Developer → Solver) and serves as the stage-game apparatus. When a solve fails, the diagnosis agent acts as an *inspector* (the Stackelberg leader) that commits, first and observably, to an inspection policy: a probing order following the pipeline’s causal dependency (data ▷ model ▷ code) together with *executed refutation* — verifying every response by a real re-solve rather than by cheap talk. The accused agents are *inspectees* (the followers): because a downstream layer is routinely implicated by an upstream defect it did not cause, each accused agent has a genuine private incentive to deflect blame and avoid wasted repair, so its interests are not aligned with truthful attribution. We show that the inspector’s committed causal-order, executed-refutation policy induces the truthful best response — a guilty layer complies and repairs, an innocent layer’s deflection is verified — replacing the baseline’s single-judge LLM accusation. We formalize the mechanism, state a proposition on causal-order inspection, and instantiate it on a *demand-uncertainty* sea-land Empty Container Repositioning (ECR) problem formulated as a two-stage stochastic linear program.

1. Introduction

This paper addresses the reliability of LLM multi-agent pipelines that automatically formulate and solve operations-research (OR) optimization problems from natural language. We inherit a working pipeline — the MAKO framework (Wu et al., 2024) — and focus on its weakest link: when the solver fails, attributing the failure to the correct pipeline stage and repairing it. We propose a state-machine Stackelberg game for this diagnosis–repair stage.

Thesis. Attributing a failed solve is not a neutral classification task but a game with conflicting interests, and the pipeline’s causal layering (*data ▷ model ▷ code*) makes it precisely a Stackelberg *inspection game*. Because an upstream defect manifests as a downstream symptom, an accused downstream agent is often innocent, and it has a genuine private incentive to deflect blame upstream and avoid the cost of a repair that could not fix the true cause anyway. The diagnosis agent is therefore an *inspector* that moves first and commits — observably — to an inspection policy: probe the layers in causal order and verify every response by *executed refutation* (a real re-solve), not by an unverifiable confidence score. This commitment is what gives the mechanism teeth: it makes truthful compliance the best response of a guilty layer and verified deflection the best response of an innocent one, so the accused can neither confess falsely nor deflect dishonestly for free. Aligning the probing order with the causal order gives the first move causal meaning — a probe of a layer upstream of the true root cause is settled by re-solve before the inspector descends. Section 2 states the demand-uncertainty ECR application model; Section 3 develops the method. Section 4 fixes the experimental protocol (Setup through Exp-I–IV, including an Exp-I pilot); numerical results and the Conclusion follow the campaign.

2. Problem Definition

We validate the diagnosis–repair mechanism on a sea–land Empty Container Repositioning (ECR) problem under *export-demand uncertainty*. The application model is a two-stage stochastic linear program on a weekly planning window (deterministic equivalent / DEP), adapted from our companion formulation for demand-uncertainty ECR.

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Unlike the MAKO baseline’s deterministic shipper–consignee multi-commodity flow, the present benchmark retains only sea and dry ports, treats empty supply as exogenous, and places all randomness in scenario-dependent export demand. Agents in the pipeline must formulate and solve this DEP from a natural-language description plus structured instance data; the expert DEP optimum supplies the strict-success rule used by the diagnosis mechanism (Section 3.4).

2.1. Network and uncertainty

The network is a directed multigraph on node set $\mathcal{N} = \mathcal{H} \sqcup \mathcal{L}$, where sea ports $\mathcal{H} = \mathcal{H}^{\text{hub}} \sqcup \mathcal{H}^{\text{spoke}}$ partition into global hubs and regional spokes, and $\mathcal{L}$ is the set of dry ports. Inland moves use modes $\kappa \in \mathcal{K}$ (road, rail). Rather than writing Big-M hinterland restrictions inside the LP, we *compile* topology into an admissible arc set

$$\mathcal{A} \subseteq \mathcal{N} \times \mathcal{N} \times \mathcal{K}, \quad (1)$$

built once from hinterland indicators λ_{ij} : include sea–dry arcs (i, j, κ) and (j, i, κ) iff $\lambda_{ij} = 1$, and dry–dry arcs (j_1, j_2, κ) iff j_1 and j_2 share a common hinterland sea port. Each arc $a = (i, j, \kappa) \in \mathcal{A}$ carries transit time τ_a , unit repositioning cost c_a^{repo} , and capacity F_a^{cap} . Flow variables exist only on $\mathcal{A}$; inadmissible hinterland pairs simply never appear.

Time is discrete by week. Empty containers held at sea or dry ports serve export demand. The empty supply $\xi_{n,t}$ at node n in week t is an *exogenous deterministic* parameter (no laden-to-empty conversion dynamics). The core uncertainty is the scenario-dependent export demand $\eta_{n,t}^\omega$ for $\omega \in \Omega$ with probabilities π_ω . Leasing is unbounded (at a premium); a heavily penalized spill variable ensures relatively complete recourse under storage caps.

A full operational system may wrap successive windows in a rolling horizon. For the agentic modeling task we fix a single window $\mathcal{T} = \{1, \dots, H\}$ (smoke instance: $H=4, |\Omega|=5$) and ask the pipeline to build the extensive-form DEP of the two-stage program below. All decisions are continuous nonnegative TEUs; at weekly scale the integrality gap is negligible.

2.2. Notation

Sets. $\mathcal{H}^{\text{hub}}, \mathcal{H}^{\text{spoke}}, \mathcal{L}; \mathcal{N}^{\text{inv}} = \mathcal{H} \cup \mathcal{L}$; modes $\mathcal{K}$; admissible arcs $\mathcal{A}$ as in (1); scenarios Ω ; window $\mathcal{T}$. For each node $n \in \mathcal{N}$ define the forward and backward stars

$$\mathcal{A}_n^+ = \{(n, j, \kappa) \in \mathcal{A}\}, \quad \mathcal{A}_n^- = \{(i, n, \kappa) \in \mathcal{A}\}. \quad (2)$$

Flow variables are indexed on arcs $(i, j, \kappa) \in \mathcal{A}$; we write $x_{ijk,t}^\omega$ interchangeably with $x_{a,t}^\omega$ for $a = (i, j, \kappa)$.

Deterministic parameters. Exogenous supply $\xi_{n,t}$; spoke top-up $E_{i,t}$; initial inventory $I_{n,0}$; committed inland arrivals $\bar{Q}_{n,t}^{\text{arr}}$ (zero on the first-window smoke instance); hub vessel calls $V_{i,t} \in \{0, 1\}$ and per-call bounds $B_i^{\text{in}}, B_i^{\text{out}}$; effective external export floor $\tilde{D}_{i,t}^{\text{ext}} = \min\{D_{i,t}^{\text{ext}}, B_i^{\text{out}}\}$; per-arc $(\tau_{ijk}, c_{ijk}^{\text{repo}}, F_{ijk}^{\text{cap}})$ for $(i, j, \kappa) \in \mathcal{A}$; storage caps U_n^{cap} ; sea costs $c^{\text{sea}, \text{in}}, c^{\text{sea}, \text{out}}$; repositioning / holding / leasing / spill costs $c_{ijk}^{\text{repo}}, c_n^{\text{hold}}, c_n^{\text{lease}}, c^{\text{spill}}$, with $c^{\text{spill}} \gg \max\{c_n^{\text{lease}}, c_n^{\text{hold}}, c_{ijk}^{\text{repo}}\}$.

Random parameters. Demand $\eta_{n,t}^\omega$ and probabilities π_ω ($\sum_\omega \pi_\omega = 1$). Scenarios are drawn by truncated-normal additive shocks about nominal means $\bar{\eta}_{n,t}$.

First-stage decisions (here-and-now). $y_{i,t}^{\text{in}}, y_{i,t}^{\text{out}} \in \mathbb{R}_+$ for $i \in \mathcal{H}^{\text{hub}}, t \in \mathcal{T}$: hub sea arrivals (arrival-week accounting) and departures, chosen *before* demand is revealed.

Second-stage decisions (wait-and-see). For each $\omega \in \Omega$: arc flows $x_{ijk,t}^\omega$ ($(i, j, \kappa) \in \mathcal{A}$), inventories $I_{n,t}^\omega$, leases $r_{n,t}^\omega$, and spill $s_{n,t}^\omega$, all in $\mathbb{R}_+$.

2.3. Two-stage formulation [D1]–[D2]

The inland submodel is a *node–arc* network formulation on the admissible digraph $(\mathcal{N}, \mathcal{A})$: topology is encoded by the stars (2), and conservation at each node is written with incidence sums over $\mathcal{A}_n^\pm$.

Stage 1 master problem [D1].

$$\min_y \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{H}^{\text{hub}}} (c^{\text{sea}, \text{in}} y_{i,t}^{\text{in}} + c^{\text{sea}, \text{out}} y_{i,t}^{\text{out}}) + \sum_{\omega \in \Omega} \pi_\omega \Theta^{\text{D}}(y, \omega) \quad (3)$$

$$\text{s.t. } y_{i,t}^{\text{in}} \leq B_i^{\text{in}} V_{i,t}, \quad \forall i \in \mathcal{H}^{\text{hub}}, t \in \mathcal{T}, \quad (4)$$

$$y_{i,t}^{\text{out}} \leq B_i^{\text{out}} V_{i,t}, \quad \forall i \in \mathcal{H}^{\text{hub}}, t \in \mathcal{T}, \quad (5)$$

$$y_{i,t}^{\text{out}} \geq \tilde{D}_{i,t}^{\text{ext}} V_{i,t}, \quad \forall i \in \mathcal{H}^{\text{hub}}, t \in \mathcal{T}, \quad (6)$$

$$y_{i,t}^{\text{in}}, y_{i,t}^{\text{out}} \geq 0. \quad (7)$$

Stage 2 recourse [D2]. Given $\mathbf{y}$ and realized η^ω , write the node–arc inflow and outflow through the stars (2) as

$$\begin{aligned} \text{In}_{n,t}^\omega &= \sum_{(i,n,\kappa) \in \mathcal{A}_n^-} x_{in\kappa,t-\tau_{in\kappa}}^\omega, \\ \text{Out}_{n,t}^\omega &= \sum_{(n,j,\kappa) \in \mathcal{A}_n^+} x_{nj\kappa,t}^\omega, \end{aligned} \quad (8)$$

with the convention $x_{ij\kappa,s}^\omega := 0$ if $s \notin \mathcal{T}$ (lagged arrivals outside the window enter via $\bar{Q}_{n,t}^{\text{arr}}$). The recourse problem is

$$\Theta^D(\mathbf{y}, \omega) = \min_{\mathbf{x}, \mathbf{I}, \mathbf{r}, \mathbf{s}} \left(C^{\text{repo}}(\omega) + C^{\text{hold}}(\omega) + C^{\text{lease}}(\omega) + C^{\text{spill}}(\omega) \right), \quad (9)$$

where the four activity cost blocks (repositioning / holding / leasing / spill) are

$$C^{\text{repo}}(\omega) = \sum_{t \in \mathcal{T}} \sum_{(i,j,\kappa) \in \mathcal{A}} c_{ij\kappa}^{\text{repo}} x_{ij\kappa,t}^\omega, \quad (10)$$

$$C^{\text{hold}}(\omega) = \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}^{\text{inv}}} c_n^{\text{hold}} I_{n,t}^\omega, \quad (11)$$

$$C^{\text{lease}}(\omega) = \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}^{\text{inv}}} c_n^{\text{lease}} r_{n,t}^\omega, \quad (12)$$

$$C^{\text{spill}}(\omega) = \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}^{\text{inv}}} c_n^{\text{spill}} s_{n,t}^\omega. \quad (13)$$

subject to

$$\begin{aligned} I_{i,t}^\omega &= I_{i,t-1}^\omega + \xi_{i,t} + r_{i,t}^\omega + y_{i,t}^{\text{in}} - y_{i,t}^{\text{out}} + \bar{Q}_{i,t}^{\text{arr}} \\ &\quad + \text{In}_{i,t}^\omega - \text{Out}_{i,t}^\omega - \eta_{i,t}^\omega - s_{i,t}^\omega, \\ \forall i \in \mathcal{H}^{\text{hub}}, t \in \mathcal{T}, \end{aligned} \quad (14)$$

$$\begin{aligned} I_{i,t}^\omega &= I_{i,t-1}^\omega + \xi_{i,t} + r_{i,t}^\omega + E_{i,t} + \bar{Q}_{i,t}^{\text{arr}} \\ &\quad + \text{In}_{i,t}^\omega - \text{Out}_{i,t}^\omega - \eta_{i,t}^\omega - s_{i,t}^\omega, \\ \forall i \in \mathcal{H}^{\text{spoke}}, t \in \mathcal{T}, \end{aligned} \quad (15)$$

$$\begin{aligned} I_{j,t}^\omega &= I_{j,t-1}^\omega + \xi_{j,t} + r_{j,t}^\omega + \bar{Q}_{j,t}^{\text{arr}} \\ &\quad + \text{In}_{j,t}^\omega - \text{Out}_{j,t}^\omega - \eta_{j,t}^\omega - s_{j,t}^\omega, \\ \forall j \in \mathcal{L}, t \in \mathcal{T}, \end{aligned} \quad (16)$$

$$0 \leq x_{ij\kappa,t}^\omega \leq F_{ij\kappa}^{\text{cap}}, \quad \forall (i,j,\kappa) \in \mathcal{A}, t \in \mathcal{T}, \quad (17)$$

$$0 \leq I_{n,t}^\omega \leq U_n^{\text{cap}}, \quad \forall n \in \mathcal{N}^{\text{inv}}, t \in \mathcal{T}, \quad (18)$$

$$r_{n,t}^\omega, s_{n,t}^\omega \geq 0, \quad \forall n \in \mathcal{N}^{\text{inv}}, t \in \mathcal{T}, \quad (19)$$

with $I_{n,0}$ fixed as the window initial inventory. Equations (14)–(16) are node–arc balances; the star sums (8) enforce hinterland feasibility through $\mathcal{A}_n^\pm \subseteq \mathcal{A}$. The remaining inequalities are arc capacities (17) and yard caps (18). Lease variables are unbounded above; spill s with large c^{spill} provides relatively complete recourse.

The agentic task is to emit a Gurobi implementation of the extensive-form DEP equivalent to (3)–(19). An expert DEP solve on the frozen smoke instance yields a known optimum z^* (approximately 1.390×10^6 for $H=4$, $|\Omega|=5$, seed 42). A pipeline run is counted as a strict success iff the solver reports OPTIMAL and the relative gap to z^* does not exceed 10^{-2} (one percent; the experimental default). Injected root-cause labels for attribution experiments are defined on the same layered pipeline (data $\triangleright$ model $\triangleright$ code) and are independent of the OR model above.

3. Proposed Method

This section develops the proposed framework bottom-up. We define the agents and their interfaces (Section 3.1), compose them into a state-machine architecture (Section 3.2), and specify their communication through a shared blackboard (Section 3.3). The final two subsections present the core contribution: a Stackelberg *inspection game* for diagnosis–repair defined *on* the state machine (Section 3.4) and its algorithmic realization (Section 3.5). Throughout, the framework targets *layered-error* natural-language optimization — OR pipelines with a data $\rightarrow$ model $\rightarrow$ code dependency in which errors propagate strictly downstream and contaminate downstream signals; we do not claim gains for general LLM reasoning.

3.1. Agent Definitions

We retain the baseline’s role-specialized agents. Let $\mathcal{A} = \{a_1, \dots, a_5\}$, with $a_1 = \text{DataEngineer}$, $a_2 = \text{ModelExpert}$, $a_3 = \text{PythonDeveloper}$, $a_4 = \text{SolverExecutor}$, $a_5 = \text{DiagnosisAgent}$. Each agent is characterized by a tuple

$$a_i = (i, \mathcal{R}_i, \mathcal{T}_i, \mathcal{B}_i, \mathcal{H}_i), \quad (20)$$

where i is a unique identifier, $\mathcal{R}_i$ the functional role, $\mathcal{T}_i$ the prompt template encoding role-specific instructions and output-format constraints, $\mathcal{B}_i$ an optional knowledge base (active only for a_2 , whose modeling decisions are domain-specialized), and $\mathcal{H}_i$ the hyper-parameter configuration (temperature, `top_p`, `max_tokens`).

The five roles decompose the modeling task along the causal order $\text{data} \triangleright \text{model} \triangleright \text{code}$. **DataEngineer** (a_1) parses the problem description and raw instance into structured parameters with symbol anchoring for downstream consistency; **ModelExpert** (a_2) designs the optimization model (decision variables, objective, constraints); **PythonDeveloper** (a_3) translates the formulation into executable solver code; **SolverExecutor** (a_4) runs the code in a sandbox, validates the result against the strict-success rule, and reports status; **DiagnosisAgent** (a_5) analyzes any failure and drives repair.

Each correctable agent exposes two maps: a *forward* map a_i^+ that produces its nominal output, and a *backward* (repair) map a_i^- that, given an error signal, either accepts responsibility and emits a corrected output or rebuts. Agent a_i receives an input assembled from prior outputs,

$$x^{(i)} \leftarrow f_i(y^{(0)}, y^{(1)}, \dots, y^{(i-1)}), \quad y^{(i)} \leftarrow a_i^+(x^{(i)}), \quad (21)$$

where f_i selects the necessary dependencies. The backward map is invoked only on the agent implicated by diagnosis (Section 3.4).

3.2. Multi-Agent Architecture and State-Machine Orchestration

The agents of Section 3.1 are composed into an extended finite-state machine rather than a free-form graph. The pipeline is compiled as

$$M = (\mathcal{Q}, q_0, \Delta, S, K), \quad (22)$$

where $\mathcal{Q}$ is a finite set of control states (the graph nodes), q_0 the unique entry state, Δ the transition map, S the shared blackboard (Section 3.3), and K the transition budget. The state set is

$$\mathcal{Q} = \{q_{\text{DE}}, q_{\text{ME}}, q_{\text{KL}}, q_{\text{PD}}, q_{\text{SE}}, q_{\text{DG}}\} \cup \{q_{\text{DE}}^-, q_{\text{ME}}^-, q_{\text{PD}}^-\}, \quad (23)$$

comprising six forward states — one per agent in $\{a_1, \dots, a_5\}$ plus a dedicated **KNOWLEDGELOADER** state — and three backward (repair) states, one per correctable agent. The entry state is $q_0 = q_{\text{DE}}$ and the lone terminal is **END**. Figure 1 depicts M .

Transitions are pure functions of the blackboard. The transition map Δ is realized by five routing functions, each a pure map $S \rightarrow \mathcal{Q}$:

- δ_{ME} : after **ModelExpert**, branch to **KNOWLEDGELOADER** iff it requests not-yet-loaded modules, else to **PythonDeveloper**;
- δ_{KL} : after **KnowledgeLoader**, return to **ModelExpert** iff new knowledge was added, else proceed to **PythonDeveloper**;

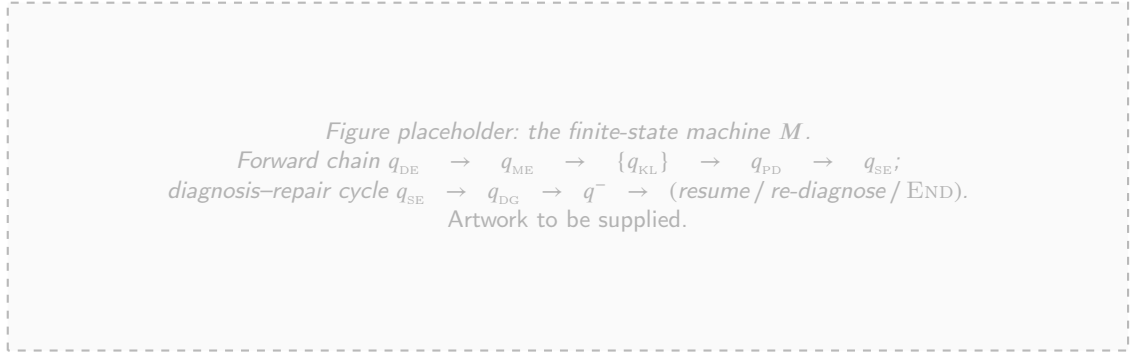


Figure 1: The extended finite-state machine $M = (\mathcal{Q}, q_0, \Delta, S, K)$. Solid arrows: the forward chain (including an optional ModelExpert knowledge-load branch). Dashed arrows: the diagnosis–repair cycle, whose return edge is determined by the Stackelberg verdict (Section 3.4).

- δ_{SE} : after SolverExecutor, terminate on success or budget exhaustion, otherwise route to DiagnosisAgent;
- δ_{DG} : after DiagnosisAgent, route to the backward state of the implicated agent (the diagnosis verdict);
- δ^- : after a backward step, either resume the pipeline downstream from the repaired agent, re-enter diagnosis, or terminate — depending on whether the repair was accepted and which agent was repaired.

The retry budget K (`max_retries`) bounds the number of diagnosis–repair cycles, i.e., it is the FSM’s transition budget over the diagnostic sub-graph.

The FSM is the stage-game apparatus, not scaffolding. The state machine is the structural substrate of the contribution because it is the object on which the repair game is defined. Its states are stage-game states, and — once the repair game is in play — the transitions δ_{DG} and δ^- become functions of the players’ strategy profile (the inspector’s committed inspection policy and the inspectee’s verified response; Section 3.4) rather than of a fixed heuristic. Treating M as the stage-game apparatus is what lets the commitment order carry *causal* meaning and what supports the proposition in Section 3.4; without this formalization, the FSM would be indistinguishable from generic graph orchestration.

3.3. Inter-Agent Communication via a Shared Blackboard

Agents never exchange messages pairwise. All communication is mediated by a single typed blackboard S (realized as the `AgentState` schema): every node reads its inputs from S and writes its outputs back to S , and the transition functions Δ route solely on S ’s contents. This *shared-state, no-direct-links* design has three consequences.

Control and payload on one schema, separated by reader. S carries both *control state* and *data payload*. Control fields — retry count, the currently implicated agent, a resolved flag, the diagnosis mode, the round counter, the retry budget — drive Δ and are the only fields a routing function inspects. Data fields — each agent’s output, the assembled error evidence, execution metrics, the loaded knowledge modules, and the ground-truth objective — carry the pipeline’s content and are read by the agent that fires next. Colocating these concerns on one schema while separating their readers is what makes the FSM’s transitions both data-aware and cheap to reason about.

A consistent, fixed data view. A deterministic preprocessor runs once, before any LLM agent fires, and writes a canonical view of the raw instance into S . Every downstream agent therefore reasons over an identical data basis, eliminating a class of data-parsing races that would otherwise masquerade as modeling or coding errors and confuse the diagnosis mechanism.

Append-by-field history. Communication is append-by-field rather than overwrite: a backward step writes a corrected output together with a resolved flag without erasing the prior forward outputs. The diagnosis mechanism can therefore inspect the full input–output history $\{(x^{(i)}, y^{(i)})\}$ when attributing a failure. This history is the evidence

base $\mathcal{E}$ on which the Stackelberg game of Section 3.4 is defined, and its completeness is what makes blame attributable rather than heuristic.

3.4. The Stackelberg Diagnosis–Repair Game as an Inspection Game

Why a game, not a single judge. The baseline diagnoses a failed solve with what it calls an “adversarial” protocol: (i) a static heuristic prior over candidate agents keyed on the solver status (CRASH $\Rightarrow$ code first; OPTIMAL-but-wrong-objective $\Rightarrow$ model first); (ii) a *single* LLM call returning a suspect, a confidence, and a rationale; (iii) a backward step in which the accused agent self-verifies. The baseline is candid that this is role-asymmetric but *not* game-theoretic — it “does not assume utility-maximizing agents, opposed payoff functions, or equilibrium behavior.” There is no payoff structure, no equilibrium concept, no leader–follower commitment, and no strategic interaction among agents.

We argue that the conflicting-interest structure is nonetheless already present and is being ignored. Under the causal layering of Section 3.4.1, an upstream defect surfaces as a downstream symptom, so the agent an observation implicates is frequently *not* the one at fault. In particular, one cannot read the root-cause layer off the SolverExecutor’s failure surface alone: a Python traceback or crash locus typically marks where the pipeline *broke*, most often in generated code, rather than where the defect was *introduced*. Treating stack traces and solver logs as a direct pointer to the guilty agent is therefore systematically biased toward the downstream code layer — precisely the single-judge failure mode that an inspection commitment is designed to avoid. A correctly self-interested accused agent should therefore *deflect* blame it does not deserve and decline a repair that cannot fix an upstream cause — exactly the incentive of an inspectee facing an inspector. We accordingly cast diagnosis–repair as a Stackelberg *inspection game* (Avenhaus et al., 2002; Maschler, 1966), a canonical class of leader–follower games in which an inspector commits to a monitoring policy and an inspectee best-responds. The diagnosis agent is the inspector (leader); the accused correctable agents are inspectees (followers); and the inspector’s first-mover *commitment* to a causal-order, executed-refutation policy is what induces truthful attribution as the inspectees’ best response. Table 1 maps the inspection-game elements to the pipeline and contrasts them with the single-judge baseline.

Table 1

Mapping of Stackelberg inspection-game elements to the diagnosis–repair stage. Right column: what the single-judge baseline provides instead.

Inspection-game element	Concretization in our mechanism	Single-judge baseline
Inspector / leader (moves first, commits)	DiagnosisAgent commits an inspection policy σ : probe order + executed refutation	One LLM call picks a suspect
Inspectees / followers (best-respond)	Accused agents a_1, a_2, a_3 : comply-and-repair or deflect-upstream	Accused agent self-checks in isolation
Committed policy σ	Causal-order probing + verify by real re-solve (no cheap talk)	Fixed heuristic prior + single confidence
Conflict of interest	Innocent layer deflects to avoid wasted repair / wrongful blame	(assumed absent)
Inspector payoff	Correct root-cause localization at fewest rounds/tokens	(undefined)
Inspectee payoff	Avoid wrongful blame and wasted repair; fix only if truly at fault	A confidence score
Induced outcome	Guilty layer complies; innocent deflection is verified	Confidence-routed re-accusation
Stage-game apparatus	The LangGraph FSM (states = stage-game states)	(treated as plumbing)

3.4.1. Layered error dependence and the deflection incentive

Let the correctable core of the pipeline be the chain $a_1 \triangleright a_2 \triangleright a_3$ (DataEngineer $\triangleright$ ModelExpert $\triangleright$ PythonDeveloper), i.e. *data* $\triangleright$ *model* $\triangleright$ *code*, where $\triangleright$ denotes “is a causal prerequisite of.” We assume *strict layered error dependence*: a defect introduced at layer $\ell \in \{1, 2, 3\}$ may manifest at any layer $j \geq \ell$, but a defect at layer ℓ cannot be the root cause of a symptom that requires only layers $< \ell$ to be well-formed. Consequently, the observation of a downstream

failure (e.g. an infeasible solve, a wrong objective, or a traceback into solver-facing code) is consistent with several root-cause layers, and localizing the true root cause ℓ^* from the failure evidence $\mathcal{E}$ alone is fundamentally ambiguous: $\mathcal{E}$ records the *symptom locus* (often a_3), not a certified identity of ℓ^* . This ambiguity is exactly what a single-judge classifier must resolve in one shot. Trusting the stack as a root-cause oracle is therefore insufficient: the failure surface motivates an *inspection* protocol rather than a one-shot classification of the traceback locus.

The conflict of interest this creates. Layered dependence does not merely make attribution ambiguous; it gives the accused agents *misaligned private incentives*, which is what turns diagnosis into a game rather than a classification. Consider an agent at layer ℓ that a failure has implicated. Two facts hold simultaneously: (i) repair is costly — a backward step re-generates the agent’s output, consumes tokens, and risks injecting a fresh defect; and (ii) if $\ell \neq \ell^*$ (the layer is innocent, merely downstream of the true cause), then *no* local repair at ℓ can make the solve pass, because by strict layered dependence the true defect lies upstream. An innocent implicated agent therefore strictly prefers to *deflect* — to rebut the accusation and point blame upstream — rather than to accept a repair that wastes effort and cannot succeed. Symmetrically, even a guilty layer, if it could avoid the repair cost by deflecting, would be tempted to do so. The accused agents are thus not neutral truth-seekers whose objective coincides with the diagnostician’s; they are self-interested inspectees. A protocol that merely *asks* an accused agent to self-verify (the baseline) cannot separate a truthful deflection from a self-serving one, because both are cheap talk. Resolving this requires an inspector that *commits* to verification, which is what Section 3.4.2 constructs.

3.4.2. Players, strategies, and the inspector’s commitment

A diagnosis–repair stage game G is instantiated whenever Δ routes from q_{SE} to q_{DG} (the solver has failed and the budget is not exhausted). The game is played on the current blackboard $S_t \in \mathcal{S}$ and has the following structure.

Players. An *inspector* I (the leader) and a set of *inspectees* (the followers). The inspector is the DiagnosisAgent a_5 , operating as an independent examiner. The inspectees are the correctable agents a_1, a_2, a_3 ; in a given probe the active inspectee is the agent $a_{\hat{\ell}}$ at the currently probed layer $\hat{\ell}$.

The inspector’s strategy is a committed policy. The inspector’s strategy is an *inspection policy* $\sigma = (\omega, \nu)$ fixed *before* any inspectee responds and observable to all: (i) a probing order ω over the layers, and (ii) a verification rule ν that resolves every inspectee response by *executed refutation* — a real re-solve under the strict-success rule — rather than by accepting the inspectee’s word. Committing to ν is the crux: it converts the inspectee’s reply from cheap talk into a claim that will be mechanically checked.

The inspectee’s strategy is a response to being probed. When probed at layer ℓ , the inspectee chooses a response $r \in \{\text{COMPLY}, \text{DEFLECT}\}$. To COMPLY is to accept responsibility and emit a *concrete repair* $h \in \mathcal{H}_{\ell}$ — a specific fix with a justification tying it to the evidence — which the inspector will execute. To DEFLECT is to rebut the accusation and assert that the root cause lies at another (upstream) layer, a claim the inspector will likewise test.

Commitment order = causal order. The probing order ω follows the causal order $a_1 \triangleright a_2 \triangleright a_3$: the inspector probes the seeded suspect layer first and, when a layer is cleared, descends to the next causal layer. Play within a probe is strictly sequential — the inspector’s policy σ is committed and observable, then the inspectee best-responds — which is what makes G a Stackelberg game. Aligning the probing order with the causal order is the central design choice: it makes the first move causally meaningful (an upstream probe is settled by re-solve before the inspector can be blamed for descending too far) rather than a coin flip among observationally equivalent layers.

3.4.3. Inspectee response and verification by executed refutation

Inspectee response. Given the evidence $\mathcal{E}$ assembled from the blackboard — stack trace, solver log, the code segment implicated by the traceback, the model definition, and the relevant data structures, together with the input–output pairs $\{(x^{(i)}, y^{(i)})\}$ of the pipeline agents — the inspectee at the probed layer $\hat{\ell}$ chooses its response

$$(r, h) \leftarrow a_{\hat{\ell}}^-(x^{(\hat{\ell})}, y^{(\hat{\ell})}, \mathcal{E}), \quad r \in \{\text{COMPLY}, \text{DEFLECT}\}, \quad (24)$$

where, if $r = \text{COMPLY}$, $h \in \mathcal{H}_{\hat{\ell}}$ is a concrete repair with a justification ρ , and if $r = \text{DEFLECT}$, h carries a rebuttal pointing at an upstream layer. Complying is a bet that “the failure is caused by my layer, and here is precisely how I fix it”; deflecting is the claim “my layer is well-formed given its inputs; the cause is upstream.”

Verification by executed refutation. Under its committed rule v , the inspector does not take the response on trust. It selects and *executes* the cheapest decisive test of the response,

$$\tau^* \leftarrow \arg \min_{\tau \in \mathcal{T}(r, h)} \text{cost}(\tau) \quad \text{s.t. } \tau \text{ is decisive}, \quad (25)$$

— a re-solve after applying h when $r = \text{COMPLY}$, or a targeted re-solve isolating the named upstream layer when $r = \text{DEFLECT}$ — and issues a verdict

$$v = (\beta, \hat{\ell}^*) = I(r, h, \tau^*, S_I), \quad (26)$$

where $\beta \in \{0, 1\}$ records whether the probed layer is *cleared* ($\beta=1$: a compliance whose re-solve still fails — a false confession — or a deflection the inspector upholds) or *confirmed* ($\beta=0$: a compliance whose re-solve passes the strict-success rule), and $\hat{\ell}^*$ is the inspector’s updated estimate of the true root-cause layer. Executing the test rather than merely “reasoning about” it is what supplies a *grounded* signal: a re-solve either meets the strict-success rule or it does not, so neither a false confession nor a dishonest deflection survives verification.

Routing from the verdict. The FSM transition δ^- is a function of the executed verdict, not of a confidence score:

- if $\beta = 0$ (confirmed), the repair $\tilde{y}$ in h is committed and Δ resumes the pipeline *downstream* of $a_{\tilde{y}}$;
- if $\beta = 1$ (cleared), Δ re-enters diagnosis and the inspector probes the next causal layer, until a confirmation or budget exhaustion (K).

This contrasts with the baseline, which routes on a single LLM confidence and re-accuses without an executed refutation.

3.4.4. Payoff structure and induced truthful attribution

Because LLMs admit no stable, observable utility function, we do *not* claim that an LLM “feels” a payoff. Instead we define numeric payoffs that the orchestrator optimizes and treat each LLM call as a *black-box best-responder*, analyzing the resulting trajectory. The inspector’s payoff rewards correct localization at low cost,

$$u_I(\sigma) = \mathbf{1}[\hat{\ell}^* = \ell^*] - \lambda_I \text{cost}(\sigma), \quad (27)$$

while the inspectee at layer ℓ , responding r , has the misaligned payoff

$$u_\ell(r; \sigma) = b \cdot \mathbf{1}[r=\text{COMPLY} \wedge \text{re-solve passes}] - c_{\text{rep}} \mathbf{1}[r=\text{COMPLY}] - c_{\text{blame}} \mathbf{1}[\text{held responsible} \wedge \ell \neq \ell^*], \quad (28)$$

where $b > 0$ is the reward for actually resolving the failure, $c_{\text{rep}} > 0$ the cost/risk of a repair, $c_{\text{blame}} > 0$ the penalty for carrying blame that does not fix the solve, $\text{cost}(\cdot)$ is token/round cost, and $\lambda_I \geq 0$ trades accuracy against budget. Eq. (28) makes the conflict explicit: the inspectee is rewarded only when it is *both* at fault *and* repairs, and is otherwise penalized for complying, so its interests diverge from the inspector’s whenever $\ell \neq \ell^*$. The ground-truth label ℓ^* enters only the *evaluation* of payoffs, never any agent’s prompt, so the game remains well-defined at inference time; equilibria are properties of the orchestrator-defined game, while each LLM is a sampled best-responder within it.

Commitment induces truthful attribution (the inspection-game deterrence effect). The value of the inspector’s first move is that its committed executed-refutation rule v reshapes the inspectee’s best response. Consider the two cases. (i) *Innocent inspectee* ($\ell \neq \ell^*$): by strict layered dependence a local repair cannot make the solve pass, so COMPLY yields $-c_{\text{rep}}$ (no reward b , and it will still be seen not to fix the solve), whereas a truthful DEFLECT avoids the repair cost and is upheld by the inspector’s targeted re-solve. Deflection strictly dominates — and, crucially, only a *truthful* deflection survives v , since a dishonest one is refuted by re-solve. (ii) *Guilty inspectee* ($\ell = \ell^*$): deflecting cannot succeed, because every upstream probe the inspector runs under causal order passes verification and the probe returns to ℓ^* having only consumed rounds; complying and repairing yields $b - c_{\text{rep}}$. Whenever $b > c_{\text{rep}}$ (a resolved solve is worth more than the repair), COMPLY is the guilty layer’s best response. Hence the committed policy makes *truthful attribution* — the guilty complies, the innocent deflects honestly — the induced best-response profile. This is the classical inspection-game deterrence result: it is the inspector’s *commitment*, not any assumed benevolence of the agents, that aligns individually optimal responses with correct root-cause localization. Without commitment (the baseline’s cheap-talk self-verification), deflection is free and truthful attribution is not incentive-compatible.

3.4.5. A proposition on causal-order inspection

Proposition 1 (Causal-order inspection lowers mis-attribution)

Under strict layered error dependence and an inspector committed to a causal-order, executed-refutation policy, each probe is *decisive*: it either confirms the probed layer (a compliance whose re-solve passes) or clears it (a false confession refuted, or a truthful deflection upheld). Consequently, causal-order inspection strictly reduces the per-round set of viable root-cause hypotheses relative to simultaneous (single-judge) play, and never eliminates the true layer. Formally, let V_{sim} and V_{ins} be the sets of root-cause layers that remain consistent with the failure observation after one round of, respectively, simultaneous diagnosis and causal-order inspection; then $V_{\text{ins}} \subseteq V_{\text{sim}}$ with $\ell^* \in V_{\text{ins}}$, and the inclusion is strict whenever a probed layer $\ell < \ell^*$ is cleared.

Sketch. Under simultaneous play the diagnostician selects a hypothesis without observing any committed, verified response, so every layer consistent with the downstream observation is a candidate; the viable set is the full observationally-equivalent upstream cone. Under committed inspection the inspectee at the probed layer ℓ must respond and the response is settled by executed refutation. If $\ell < \ell^*$, no local repair can pass the re-solve (strict layered dependence), so a compliance is refuted and a deflection upheld — either way ℓ is cleared and removed from the viable set. If $\ell = \ell^*$, the incentive analysis of Section 3.4.4 makes compliance the best response and the re-solve passes, so the true layer is confirmed and never wrongly eliminated. Causal ordering guarantees layers are cleared from the top of the upstream cone downward, so each cleared probe strictly shrinks V while simultaneous play can only guess within it. A formal bound on the mis-attribution rate and its empirical validation — centered on the commitment-order ablation — are deferred to the experimental section. $\square$

3.5. The Diagnosis–Repair Algorithm

Algorithm 1 unifies the architecture and the repair game. The FSM drives the forward pass; on failure it enters the inspection stage, where the inspector commits to a causal-order, executed-refutation policy and probes layers in causal order — each probe eliciting an inspectee response and settling it by re-solve — until a layer is confirmed or the budget K is spent. The transition map Δ is exactly the five routing functions of Section 3.2, with the return edge out of the backward state set by the executed verdict v .

Relation to the baseline. The forward chain, the blackboard, and the retry budget are inherited unchanged. The replacement is localized: the single-judge diagnosis call and its confidence-routed return are substituted by the committed inspection stage, and the return edge of δ^- is driven by the executed verdict β rather than by a confidence. This locality is deliberate — it isolates the contribution (the inspection game) from the scaffolding (the FSM) and makes the commitment-order ablation, which randomizes the order in which layers are probed, the decisive experiment: if a permuted order performs as well as causal order, the thesis is falsified.

4. Experiments

The campaign tests whether a committed causal-order inspection policy improves *verifiable* root-cause attribution under layered faults, rather than whether any strong LLM protocol can eventually repair a failed solve. Results are reported after the runs; the design below is fixed in advance.

4.1. Setup

Data. All methods use the demand-uncertainty TSLP ECR model of Section 2 on the exported dataset `prob_tslp_ecr_demand`. We start from the smoke instance `smoke_H4_Omega5` and extend to a small family of horizon/scenario configurations from the same generator. The forward pipeline, blackboard schema, strict-success rule, and sandbox solver path are shared.

Model and budget K . Provider and model are held fixed within each comparison block and reported explicitly. Every method receives the same retry budget K (`max_retries`).

Injection protocol. Ground-truth root causes $a^* \in \{a_1, a_2, a_3\}$ are obtained by controlled injection: a defect is planted at an upstream layer (DataEngineer or ModelExpert) so that the surface symptom often appears downstream (e.g., a traceback or crash locus in generated code). This matches the motivation in Section 3.4 that failure evidence records a symptom locus, not a certified identity of ℓ^* . The label a^* is used only for evaluation, never in agent prompts.

Algorithm 1 FSM-Stackelberg diagnosis–repair (inspection game)

Require: Problem $\mathcal{P}$; agents $\{a_1, a_2, a_3, a_4\}$; inspector role a_5 ; FSM $M = (\mathcal{Q}, q_0, \Delta, S, K)$; causal order $a_1 \triangleright a_2 \triangleright a_3$

Ensure: Result $Y \in \{\text{success}, \text{failure}\}$

```

1:  $S \leftarrow \text{PREPROCESS}(\mathcal{P})$ ;  $k \leftarrow 0$ ;  $Y \leftarrow \perp$ 
2: while  $Y = \perp$  do  $\triangleright$  drive the FSM
3:   // Forward chain:  $q_{\text{DE}} \rightarrow q_{\text{ME}} \rightarrow \{q_{\text{KL}}\} \rightarrow q_{\text{PD}} \rightarrow q_{\text{SE}}$ 
4:   for  $i = 1$  to 4 do
5:      $y^{(i)} \leftarrow a_i^+(S)$ ; update  $S$ ; advance  $\Delta$ 
6:   if  $y^{(4)}.\text{STATUS} = \text{OPTIMAL}$  then
7:      $Y \leftarrow \text{success}$ 
8:   else if  $k \geq K$  then
9:      $Y \leftarrow \text{failure}$ 
10:  else
11:    // Inspection stage at  $q_{\text{DG}}$ : inspector commits  $\sigma = (\text{causal order}, \text{executed refutation})$ 
12:     $\hat{\ell} \leftarrow \text{PRIOR}(y^{(4)}.\text{STATUS})$   $\triangleright$  seed probe order
13:    repeat
14:       $(r, h) \leftarrow a_{\hat{\ell}}^-(S)$   $\triangleright$  inspectee best-responds: comply-repair or deflect
15:       $\tau^* \leftarrow \arg \min_{\tau \in \mathcal{T}(r, h)} \text{cost}(\tau)$  s.t. decisive
16:       $(\beta, \hat{\ell}^*) \leftarrow a_5.\text{VERIFY}(r, h, \tau^*, S)$   $\triangleright$  executed refutation (re-solve)
17:       $k \leftarrow k + 1$ 
18:      if  $\beta = 0$  then  $\triangleright$  layer
19:         $\text{confirmedcommit } \tilde{y} \in h$ ; resume  $\Delta$  downstream
20:         $\hat{\ell} \leftarrow \text{NEXTCAUSALLAYER}(\hat{\ell})$   $\triangleright$  layer cleared: descend causal order
21:    until  $\beta = 0 \vee k \geq K$ 
22: return  $Y$ 

```

Fairness. At failure time every method reads the same blackboard (solver status, stack trace, agent input–output history). Repair uses the same agent backward_step maps and the same re-solve verification path; methods differ only in how they select or order the implicated layer. No method receives privileged tools beyond its diagnosis policy.

4.2. Methods compared

- **(A) Proposed / ablation family.** diagnosis_mode=stackelberg with committed policy $\sigma = (\omega, \nu)$ and ν fixed as executed refutation. The probing order ω is set to causal, reverse, or random.
- **(B) Internal baselines.** (i) adversarial: single-judge LLM accusation plus accused-agent self-check; (ii) sequential: fixed reverse-order backward chain without an inspection commitment.
- **(C) External verification baselines.** (i) *Debate* (Du et al., 2023): multiple diagnosis personas argue for a fixed number of rounds; terminal positions are aggregated by majority vote (no LLM-reported confidence weights; ties broken by the status prior), then the accused layer enters the shared repair/re-solve path. (ii) *Reflexion* (Shinn et al., 2023): single-judge-style accusation augmented with verbal reflections after failed diagnose–repair attempts, conditioned upon in later diagnosis rounds, again sharing the same repair/re-solve machinery.

4.3. Metrics

The primary metric is root-cause attribution accuracy $\mathbf{1}\{\hat{a} = a^*\}$, optionally stratified by injected origin agent and solver-status surface. Secondary metrics are strict-success rate at budget K (SSR@ K), realized probe rounds, and total tokens. For Exp. 4.7 we additionally record deflect rates, the fraction of compliances and deflections overturned by executed refutation, and analysis-mode episode payoffs (u_L, u_F) from Section 3.4.4. Highest SSR alone is not treated as a sufficient claim: under strong models many protocols may eventually repair; the scientific contrast is verifiable attribution under commitment.

4.4. Exp-I: Commitment-order ablation

This experiment is the decisive test of Proposition 1. Holding v and K fixed, we compare stackelberg inspection under $\omega \in \{\text{causal}, \text{reverse}, \text{random}\}$ on the injection grid of Section 4.1. Causal order is the proposed commitment; reverse and random retain commitment and multi-round probing but break alignment with data $\triangleright$ model $\triangleright$ code.

Pilot. Before the full grid we run a small upstream-fault / downstream-symptom pilot contrasting at least causal and random (or reverse) orders under identical K , optionally including a single full-context LLM attribution call as a capability reference. The pilot is interpreted as follows. If causal order yields a clear attribution gain over permuted orders, we proceed to the full ablation table. If attribution is uniformly high and causal order is indistinguishable from a permuted order, the empirical content of Proposition 1 is not supported under the present implementation or setting, and the campaign is revised before claiming a commitment-order effect. If non-committed strong attribution already saturates accuracy at lower cost, we narrow subsequent claims to mechanism and commitment-order contrasts rather than aggregate superiority. The pilot subsection may be removed once the main ablation table is in place.

4.5. Exp-II: Internal baselines

On the same injection grid and budget K we compare causal-order stackelberg inspection to the internal baselines in (B): adversarial single-judge diagnosis and sequential reverse-chain repair. We report attribution accuracy as the primary contrast, with SSR@ K and token/round cost as secondary measures of whether any attribution gain is purchased at disproportionate expense.

4.6. Exp-III: External verification baselines

We next include the external baselines in (C), multi-agent debate (Du et al., 2023) and Reflexion-style self-critique (Shinn et al., 2023), under the same blackboard, repair path, and K . The question is whether causal-order inspection retains an attribution or cost advantage once stronger multi-call LLM protocols are admitted, or whether apparent gains reduce to additional LLM invocations. Attribution is reported jointly with tokens and rounds.

4.7. Exp-IV: Inspection diagnostics

Finally we examine mechanism-level quantities predicted by Section 3.4.4: the rate at which inspectees deflect; the fraction of deflections and compliances overturned by executed refutation; and truthful-attribution rate as a function of the budget K . These diagnostics test whether the committed policy induces the intended deterrence pattern (guilty layers comply, innocent deflections are upheld only when consistent with re-solve) rather than merely adding probing rounds.

Claim boundary. Comparisons are scoped to layered-error natural-language optimization on the TSLP ECR pipeline. We do not claim gains for general LLM reasoning, nor that inspection is the unique protocol that can eventually repair a failed solve. The intended positive claim is that causal-order committed inspection improves (or cost-efficiently matches) verifiable root-cause attribution relative to unordered commitment and to non-committed verification baselines under the injection regime above.

5. Conclusion

[Placeholder] To follow once Exp-I–IV report attribution, cost, and mechanism diagnostics.

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